

Exercise Set 2

Quantitative Trading and Price Impact
MSc in Mathematics and Finance

2025-2026

- (a) **Computing price volatility and ADV.** For each trading day and symbol, compute:
- (i) the intraday price volatility, defined as the standard deviation of 10-second returns over that day;
 - (ii) the daily traded volume, defined as the total volume traded over that day.

Then, for each day and symbol, compute the corresponding 20-day trailing averages of (i) price volatility and (ii) daily traded volume, using the previous 20 trading days only (i.e., excluding the current day).

- (b) **Computing intraday impact states.** Fix a half-life $H > 0$ and define $\beta := \frac{\ln 2}{H}$. Let $(q_t)_t$ denote the signed traded volume, and let ADV and σ denote (respectively) the estimated average daily volume and volatility for a given stock (estimated from a 20-day trailing estimate as in (a)).

For a given impact model type (linear or square-root), define the normalized signed traded volume $(\tilde{q}_t)_t$ by

$$\tilde{q}_t = \begin{cases} \sigma \frac{q_t}{\text{ADV}}, & \text{(linear model),} \\ \sigma \operatorname{sgn}(q_t) \sqrt{\frac{|q_t|}{\text{ADV}}}, & \text{(square-root model).} \end{cases}$$

Assuming the impact to be zero at the start of each trading day and given $\lambda > 0$, the intraday impact state $(I_t)_{t \geq 0}$ is then defined recursively by the exponential moving average

$$I_{t+1} = (1 - \beta) I_t + \lambda \tilde{q}_t, \quad I_0 = 0,$$

or equivalently, in normalized form $(\bar{I}_t)_{t \geq 0}$ given by $\bar{I}_t := I_t / \lambda$,

$$\bar{I}_{t+1} = (1 - \beta) \bar{I}_t + \tilde{q}_t, \quad \bar{I}_0 = 0.$$

Implement a function that returns the normalized impact state series $(\bar{I}_t)_t$ for a given volume series $(q_t)_t$ and parameters (ADV, σ, H) . Finally, for half-lives $H \in \{1, 5, 30, 60\}$ minutes and both model types, compute the impact states efficiently for all symbols and all trading days.

Hint: Again, it is convenient to work with a MultiIndex `(stock, date)` with intraday time as the columns, so that you can apply an exponential weighted moving average within each `(stock, date)` pair. The pandas function `ewm` can be helpful here.

- (c) **Decomposing VWAP slippage.** Given an order flow $(Q_t)_t$, define the impact slippage by

$$\int_0^T (I_t - I_0) dQ_t,$$

where I is the intraday impact state from (b). Similarly, define the alpha slippage by

$$\int_0^T \frac{P_t - P_0}{P_0} dQ_t.$$

Assume $\lambda = 1$ and use the signed traded volume series $(q_t)_t$ from the binned sample data to define the cumulative order flow $(Q_t)_t$ by

$$Q_t := \sum_{s \leq t} q_s \quad (\text{i.e., } dQ_t = q_t).$$

Compute the corresponding impact slippage and alpha slippage for each **(stock, date)** pair.